AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Computer Science Paper II

Time: 2 hours 20 minutes Marks: 55

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Question 10 offers TWO choices. Attempt any ONE choice.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Define the term ‘software’ with reference to computer system. (1 Mark)

b. Describe FOUR stages of the data processing cycle. (4 Marks)

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Q.2. (Total 5 Marks)

There are different stages in the system development life cycle.

Draw a line to connect each stage of the system development life cycle on left side to the respective operation on right side.

Stage	Operation
Implementation	Locating and correcting code errors.
Analysis	Specifications are converted into detailed system architecture.
Debugging	Necessary changes made after the implementation of system.
Design	Installing the physical system.
Maintenance	Verifying whether the application behaves as expected or not.
	Specifying what the system is required to achieve.

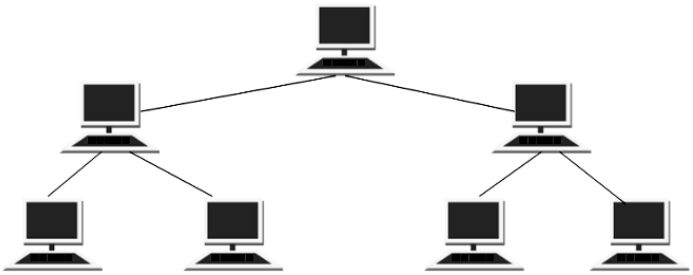
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Q.3. (Total 5 Marks)

There are different network topologies to describe the layout of devices connected on a network.

- Name the topology shown in (i).
- Write ONE advantage and ONE disadvantage of each topology in (i) and (ii).

i. (3 Marks)



Name: _____

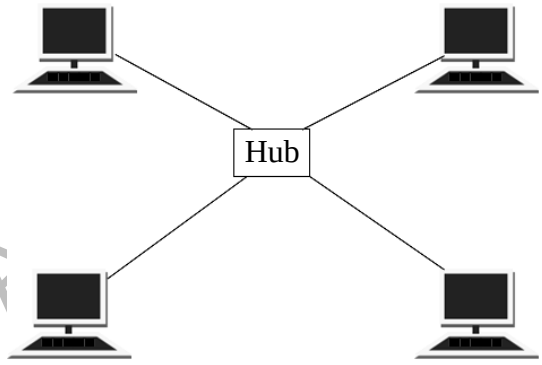
Advantage: _____

_____

Disadvantage: _____

_____

ii. (2 Marks)



Name: Star Topology

Advantage: _____

_____

Disadvantage: _____

_____

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Q.4. (Total 5 Marks)

a. Write any TWO advantages for an organisation of having a computer network. (2 Marks)

b. The open system interconnection (OSI) model is a conceptual and logical framework for communication between sender and receiver on a computer network.

i. Name the first OSI layer through which data traverse (pass) at the data sending side. (1 Mark)

ii. Name the first OSI layer through which data traverse at the data receiving side. (1 Mark)

iii. Which OSI layer decides the physical pathway between sender and receiver in a computer network? (1 Mark)

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Q.5. (Total 6 Marks)

a. Describe any THREE differences between synchronous and asynchronous transmission of data. (3 Marks)

b. Consider the given transmission of data. (3 Marks)



Identify the type of the given data transmission as serial or parallel. Write ONE benefit and ONE drawback of this type of data transmission.

Type of data transmission

Benefit

Drawback

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Q.6. (Total 5 Marks)

Computers have a number of uses in the banking sector.

Describe FIVE benefits of using computers in electronic banking.

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Q.7. (Total 5 Marks)

a. Differentiate between computer aided manufacturing (CAM) and computer aided design (CAD). (2 Marks)

b. 'Using spreadsheets, instead of calculator, paper and pen, has made the work of statisticians and accountants easier.'

Give any THREE reasons to support the given statement. (3 Marks)

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Q.8. (Total 4 Marks)

i. Define the term ‘word processor’. (1 Mark)

ii. Consider the given blank table. (3 Marks)

Write appropriate steps to create this table in MS Word.

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Q.9. (Total 5 Marks)

Consider the part of MS Excel sheet shown below.

	A	B	C	D	E	F
1	**Cool Dude Boutique**					
2	**Inventory**					
3						
4	**Item**	**Retail Value**	**Discount**	**You Pay**	**Sales Tax**	**Total Price**
5	Dress Shirt	750	140	610	36	646
6	T-Shirt	600	120	480	28	508
7	Pants	800	150	650	42	692
8	Vest	250	30	220	11	231
9	Sweater	1100	200	900	65	965

i. Write MS Excel formula to calculate the average sales tax.

(**Note:** Do not use SUM or AVERAGE functions.) (1 Mark)

ii. Write the MS Excel function with appropriate values to retrieve the highest retail value of an item. (1 Mark)

iii. Write the MS Excel function with appropriate values to retrieve the minimum total price of an item. (1 Mark)

iv. Write the MS Excel formula to count the number of items having discount value more than 120. (2 Marks)

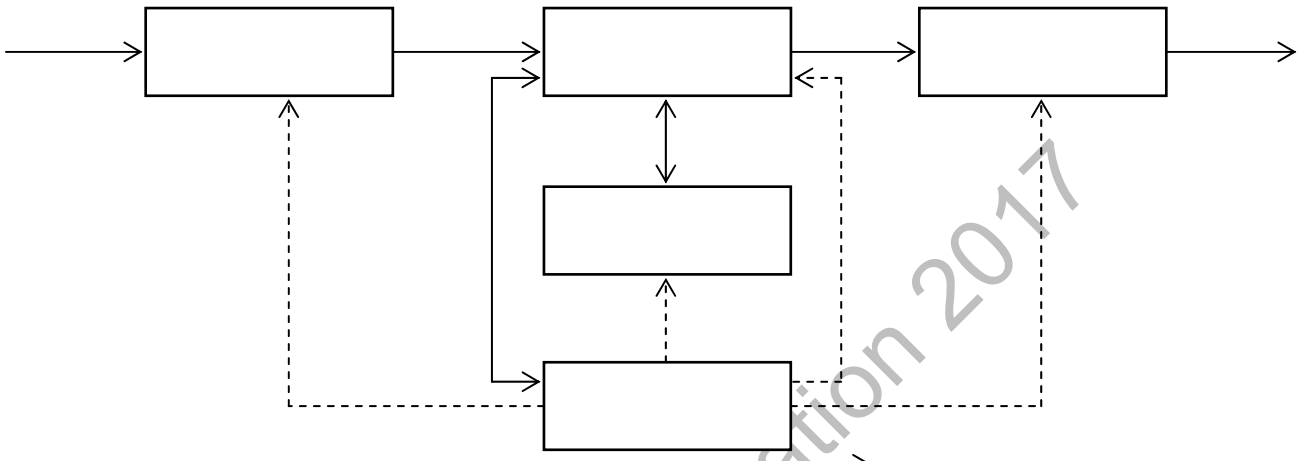
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Q.10. (Total 10 Marks)

EITHER

a.

i. Consider the blank block diagram of a computer system shown below. (5 Marks)



Note: In the given diagram, the data line is represented by ⟷ and the control line is represented by ⇢ .

Read the terms listed below and complete the above blank block diagram by placing the **number** of each term in appropriate box.

1. Control unit
2. Memory unit
3. Arithmetic logic unit
4. Register
5. Input unit
6. Output unit

ii. Describe any TWO distinct differences between random access memory (RAM) and read only memory (ROM). (2 Marks)

iii. Describe the purpose of each of the following buses in a computer system. (3 Marks)

- Data bus
- Address bus
- Control bus

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OR

b. A large number of computer users do not consider software and music piracy to be theft. More than half of computer users throughout the world admit that they pirate software according to the findings of a recent study conducted by Business Software Alliance (BSA). There is a flourishing business in music and software piracy for gaining illegal financial benefits.

i. Discuss any FIVE different methods of software and music piracy. (5 Marks)

ii. How are pirated software harmful to our systems? (2 Marks)

iii. Describe any THREE measures that should be taken to minimise piracy. (3 Marks)

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